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In an isobaric process, the pressure is constant, so P comes out of the integral,

$$W = P\int_{V_i}^{V_f} dV \qquad (8.47)$$

$$W = P[V_f - V_i] = P\Delta V \qquad (8.48)$$

Where ΔV denotes change in the volume. If ΔV is negative, W is also negative. This implies that the work is done on the gas. If ΔV is positive, W is also positive, implying that work is done by the gas.

The equation (8.48) can also be rewritten using the ideal gas equation.

From ideal gas equation

$$PV = \mu RT \text{ and } V = \frac{\mu RT}{P}$$

Substituting this in equation (8.48) we get

$$W = \mu RT_f\left(1 - \frac{T_i}{T_f}\right) \qquad (8.49)$$

In the PV diagram, area under the isobaric curve is equal to the work done in isobaric process. The shaded area in the following Figure8.36 is equal to the work done by the gas.

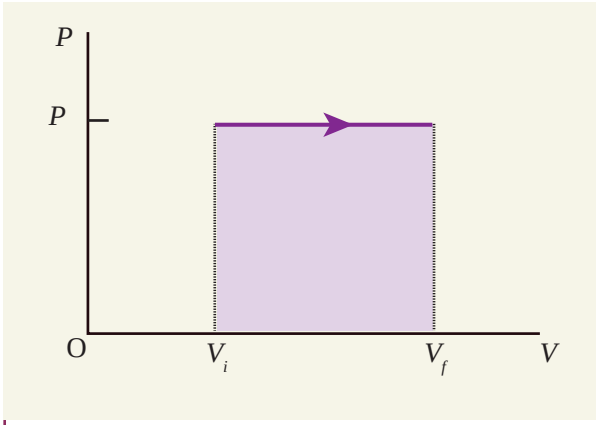


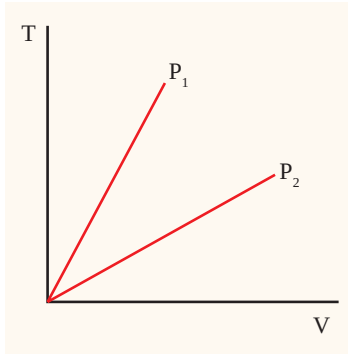
Figure 8.36 Work done in an isobaric process

The first law of thermodynamics for isobaric process is given by

$$\Delta U = Q - P\Delta V \qquad (8.50)$$

EXAMPLE 8.19

The following graph shows a V-T graph for isobaric processes at two different pressures. Identify which one occurs at higher pressure.



Solution

From the ideal gas equation, $V = \left(\frac{\mu R}{P}\right)T$

V-T graph is a straight line passing the origin.

The slope = $\frac{\mu R}{P}$

The slope of V-T graph is inversely proportional to the pressure. If the slope is greater, lower is the pressure.

Here P_1 has larger slope than P_2. So $P_2 > P_1$.

Suppose the graph is drawn between T and V (Temperature along the x-axis and Volume along the y-axis) then will we still have $P_2 > P_1$?

EXAMPLE 8.20

One mole of an ideal gas initially kept in a cylinder at pressure 1 MPa and temperature 27°C is made to expand until its volume is doubled.

(a) How much work is done if the expansion is (i) adiabatic (ii) isobaric (iii) isothermal?